DKE Numerical Mathematics 2021/2022
Exam Solutions

Remark: We use computer precision with displaying 4 dp, unless stated differently.

1. (12 points) Explain the similarity between decimal arithmetic rounded to 4 significant figures and binary double-precision floating-point arithmetic.

   Apply one step of the bisection method, followed by one step of Newton's method, to estimate a root of the function
   $$f(x) = x(x^2 - 2) - 1$$
   in the interval $[0, 2]$.
   When applying Newton's method, start in the midpoint of the interval under consideration, and use rounded arithmetic to 4 significant figures throughout your calculation.

   Solution:

   [2.5] Both are floating-point arithmetics, so have the same kind of roundoff errors, Double-precision is more accurate (16sf). Can use rounded decimal arithmetic to get insight into errors in double-precision computer arithmetic.

   [3] Take $a = 0$, $b = 2$, $c = (a + b)/2) = 1$. $f(a) = f(0) = -1$, $f(b) = f(2) = 2 \times (2^2 - 2) - 1 = 3$, $f(c) = f(1) = -2$. $\operatorname{sgn}(f(c)) = \operatorname{sgn}(f(a))$, so root in $[c, b] = [1, 2]$.

   [1] Newton method $p_{n+1} = p_n - f(p_n)/f'(p_n)$

   [1] $f'(x) = 3x^2 - 2$.

   [4.5] Take $p_0 = (1 + 2)/2 = 1.5$, $f(p_0) = -0.6250$, $f'(p_0) = 4.750$. $p_1 = p_0 - f(p_0)/f'(p_0) = 1.5 - (-0.6250)/4.750 = 1.5 - 0.1316 = 1.318$.

2. (12 points)

3. (12 points)

4. (12 points)

5. (12 points) Compute the first approximation s_1 of the discrete Fourier transform for the following $n = 6$ data points over the time interval $[0, 2\pi]$:

t_j	0	$\frac{1}{3}\pi$	$\frac{2}{3}\pi$	π	$\frac{4}{3}\pi$	$\frac{5}{3}\pi$
y_j	2.08	2.26	1.63	0.18	0.43	1.53

Hence estimate the value of y when $t = \pi/2$, and the value of $\int_0^{2\pi} f(t)^2 dt$.

Solution:

[2] Take $m = 3$, $n = 6$. $a_0 = \frac{1}{m}\sum_{j=0}^{n-1} y_j = \frac{1}{3}\sum_{j=0}^{5} y_j$, $a_1 = \frac{1}{3}\sum_{j=0}^{5} y_j \cos(t_j)$, $b_1 = \frac{1}{3}\sum_{j=0}^{5} y_j \sin(t_j)$.

[1] $a_0 = (2.08 + 2.26 + 1.63 + 0.18 + 0.43 + 1.53)/3 = 8.11/3 = 2.703$.

[2] $\cos(\pi/3) = 0.5$, $a_1 = (1 \times 2.08 + \frac{1}{2} \times 2.26 - \frac{1}{2} \times 1.63 - 1 \times 0.18 - \frac{1}{2} \times 0.43 + \frac{1}{2} \times 1.53)/3 = (2.080 + 1.130 - 0.815 - 0.180 - 0.215 + 0.765)/3 = 2.765/3 = 0.922$.

[2] $\sin(\pi/3) = \sqrt{3}/2 = 0.866$, $b_1 = 0.866 \times (0 + 2.26 + 1.63 + 0 - 0.43 - 1.53)/3 = 0.866 \times 1.93/3 = 0.557$.

[1] $s_1(t) = a_0/2 + a_1 \cos(t) + b_1 \sin(t)$.

[2] $s_1(\pi/2) = 2.703/2 + 0 \times 1.352 + 1 \times 0.557 = 1.352 + 0 + 0.557 = 1.909 = 1.91(\text{2dp})$.

[2] $\int_0^{2\pi} s_1(t)^2 dt = (a_0^2/2 + a_1^2 + b_1^2)\pi = (2.703^2/2 + 0.922^2 + 0.557^2) \times \pi = (3.654 + 0.849 + 0.310) \times \pi = 4.814 \times \pi = 15.12$.

6. (12 points)

7. (8 points) Consider the following equation for the forced Van der Pol oscillator:

$$\ddot{x} + \mu(1 - x^2)\dot{x} + x = A\cos(\omega t)$$

Show how to solve this differential equation in Matlab, including the code you would use.

Solution:

[4] Introduce variable $v = \dot{x}$. Then $\dot{v} = \ddot{x} = A\cos(\omega t) - \mu(1 - x^2)v - x = g(t, x, v)$.
State $y = (x, v)$ with $\dot{y} = (\dot{x}, \dot{v}) = (v, g(t, x, v)) = (y_2, g(t, y_1, y_2)) =: f(t, y)$.

[4] Parameters `mu=...; A=...;  omega=...;`
Acceleration `g=@(t,x,v) A*cos(omega*t)-mu*(1-x^2)*v-x;`
Right-hand side `f=@(t,y) [y(2);g(t,y(1),y(2))]`
Initial condition `x0=...; v0=...; y0=[x0;v0];`
Initial and final times `t0=0; tf=...;`
Solve system `ode45(f,[t0,tf],y0);`